

266 Natural Language Processing

Project: How to Write a Project Research Paper

This Week

Goals

- Understand the purpose of a well written NLP paper
- Review the structure of a well written project paper

Project Resources

- Valuable information in the Project Section of Syllabus
 - [Final Submission](#)
 - [Grading Rubric](#)
 - [FAQ](#)
 - [Older High Scoring Projects](#)

How to Write a Good Project Paper

Purpose

Purpose of a Paper

Clearly communicate:

- your idea,
- your intuition,
- the experiments you've designed and run,
- The results you got, and
- why you got those results

You are telling a coherent story to your colleagues.

How to Write a Good Project Paper

Structure

Paper Structure

Research papers follow a basic structure

- **Abstract**
- **Introduction** (motivation for your work)
- **Background** (literature review, or related work)
- **Methods** (description of your design and implementation)
- **Results and discussion** (tables, plots, & figures, and detailed analysis in comparison to baseline and the literature, if applicable)
- **Conclusion**
- **References**
- **Contributions** (if team of 2 or 3)
- **Appendix**

Description of Abstract

Size: 3 to 5 sentences

- Concisely describe your problem,
- how you solved it,
- and what you found

Description of Introduction

- **Size:** 1 page
- **Focus:** motivation for your work
- **Questions to answer:**
 - What is the problem you are trying to solve?
 - Why is it important?
 - Why can't it be solved without your approach?
 - What are your contributions?
 - Substantiate your refutable claims with evidence

Examples

Their Solution

In this work our aim is to identify empirically the settings that practitioners should expend effort tuning, and those that are either inconsequential with respect to performance or that seem to have a ‘best’ setting independent of the specific dataset, and provide a reasonable range for each hyperparameter. We take inspiration from previous empir-

The Problem

Unfortunately, a downside to CNN-based models – even simple ones – is that they require practitioners to specify the exact model architecture to be used and to set the accompanying hyperparameters. To the uninitiated, making such decisions can seem like something of a black art because there are many free parameters in the model. This is especially true when compared to, e.g., SVM and logistic regression. Furthermore, in practice

Other's Shortcomings

classification datasets. Most previous work in this area reports only mean accuracies calculated via cross-validation. But there is substantial variance in the performance of CNNs, even on the *same folds* and with model configuration held constant. Therefore, in our experiments we perform replications of cross-validation and report accuracy/Area Under Curve (AUC) score means and ranges over these.

Source: [A Sensitivity Analysis of \(and Practitioners' Guide to\) Convolutional Neural Nets for Sentence Classification](#)

Examples

Their Solution

This paper's key contribution is to show the surprising effectiveness of PET for text classification on small datasets related to having safe and productive online discourse (Figure 1). A novelty of our study is the scale of data we experiment with: in addition to few-shot templates that are widely studied in the LLM literature, we explore datasets too large to fit into an LLM's context window for prompting, but too small for traditional fine-tuning.² Our

The Problem

model. Training a new high-quality neural text classification model (i.e., fine-tuning models such as BERT (Devlin et al., 2019)) typically requires collecting thousands or millions of textual annotations, a process which is both time-consuming and cost-intensive.

Other's Shortcomings

Description of Background Section

- **Size:** ½ page
- **Focus:** literature review, or related work
- **Questions to answer:**
 - What have others done?
 - How were they successful?
 - How will you go beyond their success?
 - 2-3 Sentences per reference

Description of Methods Section

- **Size:** 2 - 3 pages
- **Focus:** design and implementation
- **Questions to answer:**
 - What is your proposed approach?
 - Describe your problem using an example or two.
 - What is the intuition behind your proposed approach?
 - How are you solving the problem?
 - What are your experiments/experimental design?
 - Why are the experiments helping to solve your problem?
 - How are you measuring success?

Examples

4.1 Baseline Configuration

Experimental Design

We first consider the performance of a baseline CNN configuration. Specifically, we start with the architectural decisions and hyperparameters used in previous work (Kim, 2014) and described in Table 2. To contextualize the variance in performance attributable to various architecture decisions and hyperparameter settings, it is critical to assess the variance due strictly to the parameter estimation procedure. Most prior work, unfortunately, has not reported such variance, despite a highly stochastic learning procedure. This variance is attributable to estimation via SGD, random dropout, and random weight parameter initialization. Holding all variables (including the folds)

architecture decisions and hyperparameter settings. To this end, we hold all other settings constant (as per Table 2) and vary only the component of interest. For every configuration that we consider, we replicate the experiment 10 times, where each replication again constitutes a run of 10-fold CV.⁸ We again report average CV means and associated ranges achieved over the replicated CV runs.

Examples

Experimental Design

We train soft prompts on 10, 20, 50, 80, 100, 200, 500, 1,000, and 2,000 randomly sampled training examples. For the Neutral Responses dataset, we use only up to 100 examples due to the dataset's more limited size. To account for variability in the sampling process, we repeat each experiment three times with different seeds and report average scores.

5 Baselines

5.1 In-context learning

We compare prompt-tuning to *in-context learning* (ICL) baselines, which include training data directly in the prompt-template sent to the model. We conduct experiments with zero-shot, 6-shot and 12-shot prompt-templates. For the latter two, for each seed we sample a fixed set of few-shot examples to be used for the inference prompt. For both few-shot learning and prompt-tuning, half of the training set is sampled from the positive class, half from the negative class. We repeat the few-shot experiments three times with different seeds to account for variation due to random sampling.

Description of Results and Discussion

- **Size:** 1 - 2 pages
- **Focus:** Analysis
- **Questions to answer:**
 - How well is your model performing vs. baseline?
 - What other ways might you measure its success/failure?
 - Why is your model making the errors it is?
- Back up your claims with empirical evidence
- At least one table with datasets as rows and experiments as columns

Examples

The ‘best’ number of feature maps for each filter region size depends on the dataset. However, it would seem that increasing the number of maps beyond 600 yields at best very marginal returns, and often hurts performance (likely due to overfitting). Another salient practical point is that it takes a longer time to train the model when the number of feature maps is increased. In practice, the evidence here suggests perhaps searching over a range of 100 to 600. Note that this range is only provided as a possible standard trick when one is faced with a new similar sentence classification problem; it is of course possible that in some cases more than 600 feature maps will be beneficial, but the evidence here suggests expending the effort to explore this is probably not worth it.

**Analysis:
How Well**

**Analysis:
Why**

Source: [A Sensitivity Analysis of \(and Practitioners' Guide to\) Convolutional Neural Nets for Sentence Classification](#)

Examples

Overall, these results indicate that with fewer than 100 examples, prompt-tuning PaLM 62B achieves competitive performance across attributes, demonstrating its ability to serve as a method for efficiently building safety classifiers in data-scarce settings.

6.3 Neutral Responses

Analysis: How Well

Our results show that prompt-tuning on small datasets consistently achieves performance that is competitive with the previous state-of-the-art (e.g., BERT-based fine-tuning approaches that use much larger datasets of human-annotated examples). Prompt-tuning performance also appears to be equal to or better than human-annotation quality. We found that ICL with few-shot templates is sometimes very effective, but also has much more variable performance.

Table in Results and Discussion

Your experiments can be summarized as a table with

Data variants as rows

Experiments as columns

Dataset	Experiment 1	Experiment 2	Experiment 3
Data set balanced			
Data set unbalanced			
Data set two			

Results and Discussion Table Example

Model	MR	SST-1	SST-2	Subj	TREC	CR	MPQA
CNN-rand	76.1	45.0	82.7	89.6	91.2	79.8	83.4
CNN-static	81.0	45.5	86.8	93.0	92.8	84.7	89.6
CNN-non-static	81.5	48.0	87.2	93.4	93.6	84.3	89.5
CNN-multichannel	81.1	47.4	88.1	93.2	92.2	85.0	89.4

Four experiments on embeddings

Seven different datasets

From Kim (2014) [Convolutional Neural Networks for Sentence Classification](#)

Results and Discussion Table Example

Model	Dialogue Safety					Neutral Responses		
	PARLAI SINGLE STANDARD	PARLAI SINGLE ADVERSARIAL	PARLAI MULTI	BAD-2	BAD-4	Multiple Perspectives	Neutral	Well- Explained
PaLM 62B best few-shot	0.89	0.67	0.56	0.54	0.54	0.84	0.87	0.87
T5 XXL - 80	0.18	0.18	0.19	0.29	0.48	0.94	0.96	0.76
T5 XXL - 2,000	0.90	0.91	0.48	0.20	0.44	—	—	—
Human Agreement	—	—	—	—	—	0.94	0.95	0.90
Previous SOTA	0.88	0.67	0.66	—	—	—	—	—
PaLM 62B - 80	0.87	0.77	0.71	0.60	0.65	0.94	0.96	0.88
PaLM 62B - 2,000	0.95	0.91	0.81	0.68	0.70	—	—	—

Five experiments on soft prompt tuning

Three datasets

Source: [Towards Agile Text Classifiers for Everyone](#)

Description of Conclusion

- **Size:** 3 to 5 sentences
- **Questions to answer:**
 - What is your problem?
 - Did you solve it?
 - What did you find?
 - Future work (optional)

Project Contributions

If you are in a group of 2 or 3:

In your final paper, you **MUST** include an “Authors Contributions” section where you provide the name of each team member and the contributions made by each team member to the project

Authors' Contributions Example

Authors' Contributions

AB, MB, and BA were involved in the conceptualization of the project. AB and MB developed the experimental design. BA did the initial data analysis. AB built the baseline model and fine-tuned the BERT model. MB fine-tuned the RoBERTa model, and BA fine-tuned the DeBERTa model. MB and BA performed the analysis and AB prepared the draft manuscript. MB and BA reviewed and edited the manuscript. All authors read and approved the final manuscript.

Description of References and Appendix

- References
 - We want to see that you've consulted the literature
- Gen AI Usage Policy
 - “minimally cite uses of GenAI in either the References or other designated section by indicating which model you used and the task you used it for.”
- Appendix
 - 4 to 6 page limit so put extra results, data samples, etc. in Appendix

How to Write a Good Project Paper

Questions?